Kaneshwar Sharma

Senior Software Engineer | *Python & FastAPI* | *Go* | *MySQL & PostgreSQL* | *Microservices* | *AWS & GCP*
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Summary

Backend-focused senior software engineer who builds Python (FastAPI) and Go REST APIs, event-driven microservices, and data pipelines on Linux, AWS, and GCP. At Spyne.ai, was promoted to SDE-II and team lead of 3 within 13 months, and shipped FastAPI APIs with API-key authentication for **50+ enterprise clients**, JWT authentication for its web and mobile apps, and a Kafka pipeline processing **200K+ images/day**. At Intract, a Web3 platform, built the backend for a multi-chain analytics launch that generated **$600K+ in NFT mint revenue within 2 weeks**. Writes unit tests against mocked dependencies (pytest, unittest.mock) behind CI coverage gates; also works in TypeScript, with early-career production React/Next.js experience.

Work Experience

Turing — Jul 2025 – Jun 2026

Senior Backend Engineer

AI model-training work and a client placement with a Dubai-based company building a super app

Stack: Go, PostgreSQL (client placement)

- Developed Go and PostgreSQL backend services integrating Uber into the client's super app (OAuth 2.0 account linking, fare estimates, and trip-status webhooks), so users could access Uber rides without leaving the app.
- Built backend APIs that create standardized AI/ML training environments, with consistent data formatting and scalable ingestion pipelines.

Intract — Jan 2024 – Jul 2025

Senior Backend Engineer (SDE-III) – consultant, converted to full-time

Web3 platform: multi-chain user analytics and wallet intelligence

Stack: Python, TypeScript/Node.js, Go · GCP, AWS · MongoDB, Redis, ChromaDB · Web3: MetaMask, on-chain data

- Built the backend for Intract Rewind end-to-end: parallel fetchers plus a normalization/aggregation layer over **16 blockchain data sources** that reconstructed each user's on-chain activity history behind an analytics API; the launch generated **$600K+ in NFT mint revenue within 2 weeks**.
- Led the AWS-to-GCP migration, building the migration tooling in Python, and a multi-region rollout on autoscaling Managed Instance Groups behind a Global HTTP(S) Load Balancer; database indexing, query restructuring, caching, gzip compression, and instance right-sizing cut p95 response times **50%** and monthly cloud spend **40%**.
- Built the serverless Go backend (API Gateway → Lambda → Kinesis stream processors → MongoDB) for a Telegram Mini App game serving ∼**5M users (40K+ daily active)**, with unit and integration tests run in Docker on GitHub Actions CI/CD.
- Shipped a Web3 AI assistant turning MetaMask wallet activity, social profiles, and on-chain data into personalized insights: 5+ intent-routed agents, ChromaDB vector search with per-user data isolation, and SSE streaming.
- Implemented JWT authentication with access and refresh tokens (cookie-based) for the Intract web app's backend APIs.
- Automated infrastructure provisioning with Terraform and a Docker registry, with secrets managed in HashiCorp Vault.

Spyne.ai — Mar 2021 – Jan 2024

SDE-I → Senior Software Engineer (SDE-II) / Team Lead — *Gurugram, Haryana*

AI image-processing SaaS for automotive dealers and e-commerce platforms

Stack: Python, FastAPI, Kafka, SQS · MySQL (SQLAlchemy, Alembic), MongoDB, Redis · AWS EKS, Docker · Prometheus, Grafana

- Designed and maintained Python/FastAPI REST APIs for **50+ enterprise clients**, with versioned endpoints and OpenAPI docs, and built their API-key authentication end-to-end.
- Worked directly with enterprise clients, product managers, designers, and customer success to understand integration requirements, onboard clients onto the APIs, and resolve integration issues.
- Promoted from SDE-I to SDE-II and **team lead of 3** within 13 months (Apr 2022): conducted code reviews, mentored new hires, and led API architecture and MySQL schema design (Alembic migrations).
- Engineered an event-driven microservices pipeline on Kafka (confluent-kafka) and AWS SQS processing **200K+ images/day**, using Python multiprocessing to run pipeline workers in parallel across CPU cores.
- Built a custom autoscaling controller that scaled worker pods on AWS EKS by Kafka consumer lag, throughput, and processing latency, cutting infrastructure cost **35%**.
- Added a Redis caching layer that reduced database load **60%** and API latency from **500ms to under 100ms**; set up Prometheus/Grafana monitoring and alerting for incident response.
- Delivered the backend for Spyne's Android/iOS apps with real-time processing-status updates, and JWT authentication (access + refresh tokens) for its web and mobile apps.

Appsuccessor **Sep 2020 – Mar 2021**

Software Engineer

– Automated replication and monitoring of third-party API data streams, improving the reliability of marketing analytics.
– Developed REST API integrations and web-scraping automation that reduced manual data collection and human error.

PROJECTS

- **Browser-Automation Engine** *(Python, asyncio, FastAPI, Playwright, React, TypeScript)*: automates multi-step job-application forms with async Playwright; FastAPI runs workflows as BackgroundTasks and streams progress over Server-Sent Events (async generator with asyncio.Queue fan-out) to a React + TypeScript operator UI; **136 pytest tests** using pytest-asyncio and AsyncMock.
- **AI Web3 Assistant** *(RAG, ChromaDB, embeddings, LLM agents)*: agentic AI platform using RAG, ChromaDB vector embeddings, and custom LLM agents to generate personalized blockchain insights from wallet activity, social profiles, and on-chain behavior.
- **Blockchain Analytics Platform**: large-scale data pipelines aggregating and normalizing activity across 16 blockchain data sources, powering user analytics, engagement scoring, and historical journey reconstruction.
- **Real-Time WebSocket Gateway** *(Node.js, TypeScript, Redis Pub/Sub)*: horizontally scaled WebSocket servers without sticky sessions, using Redis Pub/Sub to broadcast events and stream chunked LLM responses to clients on any instance.

TECHNICAL SKILLS

Languages: Python, Go, TypeScript, JavaScript, SQL
Backend: FastAPI, Django, Pydantic, SQLAlchemy, asyncio, multiprocessing, Node.js/Express (TypeScript), REST APIs, OpenAPI, SSE, WebSockets; familiar with Flask
Go: Gin, net/http, goroutines and channels, context, AWS Lambda (Go runtime), go test
Auth & Security: JWT authentication (access + refresh tokens, cookie-based) for web and mobile apps, API-key authentication for B2B integrations, HashiCorp Vault secrets management
Databases: MySQL, PostgreSQL (schema design, indexing, transactions, Alembic migrations), MongoDB, Redis
Testing & Observability: pytest (fixtures, parametrize, monkeypatch, pytest-asyncio), unittest.mock (patch, MagicMock, AsyncMock), FastAPI TestClient with dependency_overrides, go test, Vitest, Artillery load testing, Postman; coverage gates in CI; Sentry, Prometheus, Grafana
Architecture: Microservices, event-driven systems (Kafka, SQS, Kinesis, Redis Pub/Sub), autoscaling, caching
Cloud & DevOps: Linux (shell scripting, debugging), AWS (Lambda, API Gateway, Kinesis, SQS, RDS), GCP (Compute Engine, Cloud Load Balancing, Cloud SQL, Pub/Sub, Cloud Run), Docker, Kubernetes (EKS), Terraform, GitHub Actions (CI/CD), Nginx
Frontend: React, Next.js, TypeScript (production UI early in career; used again in recent projects)
AI/LLM: RAG, embeddings, vector search (ChromaDB), multi-agent orchestration, LLM streaming, Vercel AI SDK
Web3 (Intract): multi-chain blockchain data aggregation (16 sources), on-chain analytics and user-journey reconstruction, wallet intelligence, MetaMask (EVM wallet) integration, NFT mint launches, Web3 AI agents

EDUCATION

Guru Gobind Singh Indraprastha University **2016 – 2020**
Bachelor of Technology, Computer Science and Engineering *Delhi, India*